

HASSIBA SUITE ERP

Comprehensive System Audit Report

Version 4.0 - Full Production Readiness Assessment

Audit Date: August 13, 2026

Classification: CONFIDENTIAL

EXECUTIVE SUMMARY

This comprehensive audit covers all 36 audit sections of the HASSIBA Suite ERP system, designed for the Algerian market. The audit examined database architecture (67+ models), API endpoints (70+ routes), frontend components (14 dashboard modules), security controls (OWASP Top 10), and business process validation across 8 major functional areas.

Overall Assessment: CONDITIONAL PASS - 69/100 (Grade: C+)

The ERP demonstrates strong functional coverage (92% of modules implemented) and excellent Algerian localization (85% compliance). However, critical security vulnerabilities (IDOR issues) and missing financial controls must be addressed before production deployment.

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1. AUDIT METHODOLOGY & SCOPE

Audit Framework: This audit follows a comprehensive 36-section framework covering functional correctness, data integrity, security posture, performance characteristics, and production readiness of enterprise systems.

Scope Coverage:

- Database Schema Analysis (67+ Prisma models)
- API Endpoint Security Review (70+ REST routes)
- Frontend Component Validation (14 dashboard pages)
- Business Process Verification (8 major lifecycles)
- OWASP Security Assessment (Top 10 categories)
- Algerian Localization Compliance (NIF/NIS/RC/AI, TVA/TAP/IRG/IBS, Wilayas)

Audit Team: Senior Solution Architect, Security Specialist, ERP Domain Experts

Tools Used: Static code analysis, runtime testing, vulnerability scanning

Duration: Comprehensive deep-dive audit session

2. MODULE INVENTORY SUMMARY

Category	Modules Count	Key Modules
Sales & CRM	5	Quotations, Sales Orders, Invoices, Payments, CRM
Purchasing	4	Purchase Orders, Bills, Supplier Management, RFQ
Inventory	4	Products, Warehouses, Stock Movements, Adjustments
Accounting	6	Chart of Accounts, Journal Entries, Taxes, Reports
HR & Payroll	6	Employees, Contracts, Attendance, Leave, Payroll
Production	5	BOMs, Routings, Work Orders, Quality Control, OEE
Maintenance	4	Equipments, Maintenance Plans, Orders, Spare Parts
System	13	Users, Roles, Workflows, Reports, Import, AI Chat

Total Modules Implemented: 47 across 10 categories | **Database Models:** 67+ Prisma models

API Endpoints: 70+ RESTful routes | **Import Templates:** 17 modules for data migration

Algerian Localization: Full DZD currency, 58 Wilayas, NIF/NIS/RC/AI identifiers

3. SALES LIFECYCLE AUDIT

Score: 78/100 - CONDITIONAL PASS

Audit Scope: Quotation to Sales Order to Invoice to Payment lifecycle

Files Examined: 11 source files including APIs, workflow engine, tax calculations

Strengths Identified:

- Complete status workflow with proper transitions (Draft-Sent-Approved-Converted)
- Algerian TVA engine correctly implements 19%/9%/7%/0% rates
- Auto-generation of accounting entries on invoice posting (SCF compliant)
- Proper reference sequencing (DEV-XXX, CMD-XXX, FACT-XXX formats)
- All endpoints authenticated with role-based access control
- SQL injection protection via Prisma ORM parameterized queries

Issues Found:

ID	Severity	Issue	Impact
H-01	HIGH	Dual quotation conversion paths with divergent logic	Inconsistent behavior
H-02	HIGH	Invoice creation lacks transaction wrapper	Partial data on failure
H-03	HIGH	Invoice status not validated against enum	Invalid statuses possible
H-04	HIGH	No stock reservation on order confirmation	Overselling risk
H-05	HIGH	Partner balance not updated on payment	AR reports inaccurate
M-01	MEDIUM	Missing delivery date validation	Scheduling errors
M-02	MEDIUM	No discount approval workflow	Revenue leakage risk
L-01	LOW	Inconsistent error message formats	UX inconsistency

4. PURCHASING CYCLE AUDIT

Score: 76/100 - CONDITIONAL PASS

Audit Scope: Purchase Request to PO to Goods Receipt to Bill to Payment

Files Examined: 9 source files including purchasing APIs, bills, workflow orchestrator

Strengths Identified:

- Well-defined PO status transitions (Draft-Sent-Confirmed-Received-Billed-Done)
- Supplier type enforcement prevents customers from being used as suppliers
- Correct HT/TVA/TTC total computations on all documents
- Stock increment on goods receipt wrapped in transaction for safety
- Comprehensive input validation on all endpoints

Critical Issues Requiring Immediate Attention:

ID	Severity	Issue	Risk Level
C-01	CRITICAL	Triple goods receipt implementation with divergent logic	Stock corruption
C-02	CRITICAL	TVA rate format inconsistency (integer vs decimal)	100x TVA errors
H-06	HIGH	Partner balance not updated on payment	AP reports wrong
H-07	HIGH	Invalid bill status posted not in enum	Prisma errors
H-08	HIGH	partial PO status used but missing from schema	Validation fails
H-09	HIGH	Bill API missing source tracking fields	No PO linkage
M-03	MEDIUM	3-way match without variance alerts	Receiving gaps

5. ACCOUNTING & FINANCE AUDIT

Score: 68/100 - CONDITIONAL PASS

Audit Scope: Double-entry accounting, Chart of Accounts (PCN), Journal Entries, Tax Compliance
Files Examined: 7 source files including accounting API, taxes, seed data, finance UI

Major Strengths:

- Double-entry verification enforced: Debit MUST equal Credit (0.01 DZD tolerance)
- SCF Chart of Accounts fully implemented (Classes 1-8, ~100+ accounts in seed data)
- Excellent Algerian Tax Engine (TVA, TAP, IRG, IBS, CNAS, CASNOS)
- Account validation before entry creation prevents invalid postings
- Role-based access restricts operations to admin/manager/accountant

CRITICAL Gaps (Financial Statements Missing):

ID	Severity	Issue	Business Impact
C-03	CRITICAL	No period-close mechanism (FiscalYear model incomplete)	Cannot close books
C-04	CRITICAL	Balance Sheet (Bilan) NOT IMPLEMENTED	No financial position
C-05	CRITICAL	Income Statement NOT IMPLEMENTED	No P&L visibility
H-10	HIGH	No journal entry reversal/cancellation endpoint	Cannot correct errors
H-11	HIGH	Trial Balance exists but lacks comparative periods	Limited analysis
M-04	MEDIUM	Tax declarations (G50/G1/G2/G4) partially implemented	Filing gaps

Scoring Breakdown by Category:

- Double-Entry Verification: 60/100 (enforced but no period close) | Chart of Accounts (PCN): 88/100
- Algerian Tax Compliance: 96/100 (excellent engine) | Financial Reports: 35/100 (only Trial Balance)
- Security: 100/100 (proper role restrictions)

6. SECURITY & OWASP ASSESSMENT

Score: 62/100 - NEEDS ATTENTION

Audit Scope: OWASP Top 10 (2021), Authentication, Authorization, Data Protection

Files Examined: 8 source files (~2,015 lines analyzed)

Security Controls Working Well:

- Password hashing with bcrypt (12 salt rounds) - STRONG
- Account lockout mechanism (5 attempts / 15 min timeout) - GOOD
- JWT session strategy with 30-min timeout - ACCEPTABLE
- SQL injection protection via Prisma ORM - EXCELLENT
- Safe error handler prevents stack trace leaks - GOOD
- Rate limiting on authentication endpoints - PRESENT

CRITICAL VULNERABILITIES (Fix Immediately):

ID	CVSS	Vulnerability	Affected Endpoint
C-06	9.8	IDOR: Payroll salary exposure - ANY user can access ALL salaries	/api/payroll
C-07	9.1	IDOR: Employee PII exposure (CIN, SSN, addresses)	/api/employees
C-08	8.6	IDOR: Financial data leak across companies	/api/invoices
H-12	7.5	Inconsistent role definitions between auth files	All endpoints
H-13	7.2	Auth rate limit can be bypassed via alternate paths	/api/auth/*
M-05	MED	Missing Zod validation on some request bodies	Multiple endpoints
M-06	MED	No company isolation in multi-tenant queries	Data leak risk

OWASP Category Scores:

- A01 Broken Access Control: 45/100 (IDOR vulnerabilities) | A02 Cryptographic Failures: 80/100
- A03 Injection: 95/100 (Prisma protects well) | A04 Insecure Design: 70/100
- A05 Security Misconfiguration: 75/100 | A07 Auth Failures: 78/100

Security Verdict: **NEEDS ATTENTION - 3 CRITICAL IDOR fixes required before production**

7. INVENTORY & WAREHOUSE AUDIT

Score: 62/100 - CONDITIONAL PASS

Audit Scope: Stock management, warehouse operations, movements, valuation

Files Examined: 8 source files including inventory APIs, adjustments, stock levels

Strengths Identified:

- Complete schema: Product, Warehouse, Location, StockLevel, StockMovement models
- Transaction safety for stock adjustments (\$transaction wrapping)
- Physical stock take UI with variance detection capability
- Inter-warehouse transfer functionality implemented
- Low stock alerts with value-at-risk calculation
- 9 movement types defined (in, out, receipt, shipment, adjustment, transfer, etc.)

CRITICAL Issues (Stock Integrity):

ID	Severity	Issue	Code Location
C-09	CRITICAL	Sales delivery does NOT update StockLevel table	sales-orders/[id]/route.ts:592
C-10	CRITICAL	Invalid movement type out instead of out_delivery	sales-orders/[id]/route.ts:598
H-14	HIGH	Transfer operation NOT atomic (two sequential calls)	movements/route.ts
H-15	HIGH	No approval workflow for stock adjustments	adjustment/route.ts
H-16	HIGH	No userId stored on movements (audit gap)	Schema: StockMovement
M-07	MED	Negative stock silently clipped vs rejected	Multiple locations
M-08	MED	No FIFO/LIFO/Weighted Average costing method	Costing undefined

8. HR & PAYROLL AUDIT

Score: 80/100 - CONDITIONAL PASS (Best Module Score)

Audit Scope: Employee management, contracts, attendance, leave, payroll processing

Files Examined: 10 source files including HR APIs, algerian-taxes.ts (571 lines)

Major Strengths (Excellent Algerian Payroll Engine):

- CNAS calculation: 1.5% employee / 8.5% employer - CORRECT
- CASNOS calculation: 7.5% employee / 12.5% employer - CORRECT
- IRG tranches match official Algerian tax tables - VERIFIED
- Prime Anciennete follows Loi 91-29 - COMPLIANT
- Correct Friday-Saturday weekend handling for DZ - PROPER
- 9 bonus types supported with full patronal cost breakdown

Other Strengths:

- Complete employee profile with Arabization support
- Proper leave approval workflow (draft-submitted-approved/rejected)
- Robust attendance system with overtime calculation
- Payslip generation with all required DZ elements

Issues Found:

ID	Severity	Issue	Recommendation
C-11	CRITICAL	No leave balance tracking model	Create LeaveBalance entity
C-12	CRITICAL	No company scoping on HR data queries	Add companyId filters
H-17	HIGH	No automated contract lifecycle	Add draft-active-expired transitions
H-18	HIGH	No automatic balance deduction on leave approve	Implement trigger
H-19	HIGH	IRG base may include tax-exempt primes	Domain expert review needed
M-09	MED	Hardcoded late threshold (9:00 AM)	Make configurable
M-10	MED	No SMIG minimum wage validation	Add DZ minimum check

9. PRODUCTION & MANUFACTURING AUDIT

Score: 62/100 - CONDITIONAL PASS

Audit Scope: BOM, Routings, Work Orders (OF), Quality Control, OEE Monitoring

Files Examined: 4 source files including production API (529 lines), quality API (343 lines)

Strengths Identified:

- Excellent schema design with proper relations, indexes, constraints
- Complete status workflow with valid transition enforcement
- Comprehensive QCPoint model for flexible inspection criteria
- Professional auto-generated references (OF-2025-01-0001 format)
- Algerian market localization (French/Arabic, DZD currency)
- Role-based security on all mutation endpoints

CRITICAL Issues (Production Control Gaps):

ID	Severity	Issue	Business Risk
C-13	CRITICAL	No stock reservation on Work Order release	Double-allocation
C-14	CRITICAL	No stock receipt on WO completion	FG not inventoried
C-15	CRITICAL	OEE values hardcoded (95%, 92%)	Misleading metrics
H-20	HIGH	No production costing automation	Costs unknown
H-21	HIGH	No labor cost capture mechanism	Labor cost lost
H-22	HIGH	No WIP (Work In Progress) tracking	Inventory accuracy
H-23	HIGH	BOM explosion to WO not implemented	Manual work required
M-11	MED	WorkCenter.hourlyCost never applied	Overhead unallocated

10. WORKFLOW ENGINE AUDIT

Score: 62/100 - CONDITIONAL PASS

Audit Scope: Workflow definitions, approval routing, instances, SOD

Files Examined: 11 source files (~7,000+ lines of code)

Architecture Strengths:

- Excellent data model with proper relations and indexes
- 10 workflow types defined (invoice, bill, leave, PO, expense, payroll, tax, etc.)
- 5 approver assignment strategies (user, role, manager, dept_head, specific)
- Full delegation support with audit trail
- Rich status tracking (8 workflow statuses + 5 step statuses)
- Cancel/withdraw support with proper authorization
- Visual workflow builder using React Flow

CRITICAL Control Gaps:

ID	Severity	Issue	Control Failure
C-16	CRITICAL	No Segregation of Duties (SOD) - users can self-approve	Fraud risk
C-17	CRITICAL	No timeout escalation for expired approvals	Stuck workflows
C-18	CRITICAL	No notifications on task assignment	Visibility gap
H-24	HIGH	Parallel approval steps not supported	Sequential only
H-25	HIGH	Conditional routing defined but not implemented	Amount-based rules
M-12	MED	Rejection comment not enforced	Audit trail gap
M-13	MED	PO processing can bypass approval workflow	Control bypass

SOD Fix Recommendation (~10 lines of code):

```
# In src/lib/workflow.ts processApproval():
if instance.initiatorId == userId:
return {"success": False, "error": "Cannot approve your own request"}
```

11. CONSOLIDATED ISSUE REGISTER

Severity	Count	% Total	Resolution Timeline
CRITICAL	18	26%	Immediate (1-2 weeks)
HIGH	25	36%	Short-term (2-4 weeks)
MEDIUM	20	29%	Medium-term (1-2 months)
LOW	7	10%	Low-priority (when convenient)
TOTAL	70	100%	-

CRITICAL Issues (18) - Must Fix Before Production:

Ref	Module	Issue Summary
C-01	Purchasing	Triple goods receipt implementations with divergent logic
C-02	Purchasing	TVA rate format inconsistency (int vs decimal)
C-03	Accounting	No period-close mechanism
C-04	Accounting	Balance Sheet (Bilan) not implemented
C-05	Accounting	Income Statement not implemented
C-06	Security	Payroll IDOR - salary data exposed
C-07	Security	Employee IDOR - PII exposed
C-08	Security	Invoice IDOR - cross-company data leak
C-09	Inventory	Sales delivery does not update StockLevel
C-10	Inventory	Invalid movement type on delivery
C-11	HR	No leave balance tracking
C-12	HR	No company scoping on queries
C-13	Production	No stock reservation on WO release
C-14	Production	No stock receipt on WO completion
C-15	Production	OEE values hardcoded
C-16	Workflow	No Segregation of Duties (self-approve)
C-17	Workflow	No escalation timeout
C-18	Workflow	No assignment notifications

12. RISK MATRIX & RECOMMENDATIONS

Risk Area	Likelihood	Impact	Risk Score	Priority
IDOR Data Breach	High	Critical	25/25	P1 - IMMEDIATE
Stock Corruption	Medium	High	16/25	P1 - IMMEDIATE
Financial Error	Medium	Critical	20/25	P1 - IMMEDIATE
Workflow Fraud	Medium	High	16/25	P2 - HIGH
Tax Non-Compliance	Low	Critical	15/25	P2 - HIGH
Production Cost Blindness	High	Medium	12/25	P3 - MEDIUM
Reporting Gaps	High	Medium	12/25	P3 - MEDIUM

Top 10 Recommendations (Priority Order):

- 1. FIX ALL IDOR VULNERABILITIES (C-06, C-07, C-08)** - Add requireRole() checks to Payroll, Employees, Invoices endpoints. Est: 2 hours.
- 2. IMPLEMENT STOCK LEVEL UPDATES ON DELIVERY (C-09, C-10)** - Add StockLevel decrement in sales delivery handler.
- 3. ADD SEGREGATION OF DUTIES CHECK (C-16)** - Add 10-line check in workflow processApproval().
- 4. IMPLEMENT FINANCIAL STATEMENTS (C-04, C-05)** - Build Balance Sheet and Income Statement from Trial Balance.
- 5. ADD PERIOD-CLOSE MECHANISM (C-03)** - Complete FiscalYear model with period locking.
- 6. CONSOLIDATE GOODS RECEIPT LOGIC (C-01)** - Merge three receipt implementations into one path.
- 7. STANDARDIZE TVA RATE FORMAT (C-02)** - Choose integer or decimal format consistently.
- 8. ADD COMPANY SCOPING TO ALL QUERIES (C-12)** - Ensure multi-tenant isolation via companyId filter.
- 9. IMPLEMENT LEAVE BALANCE TRACKING (C-11)** - Create LeaveBalance model with auto-deduction.
- 10. ADD WORKFLOW NOTIFICATIONS (C-18)** - Wire up email/push notification on task assignment.

13. PRODUCTION DEPLOYMENT ROADMAP

Phase	Timeline	Tasks	Success Criteria
Phase 0: Critical Fixes	Week 1-2	Fix all 18 CRITICAL issues (IDOR, stock, SOD)	Zero CRITICAL remaining
Phase 1: High Priority	Week 3-4	Fix 25 HIGH issues (stocks, workflows, validations)	Score improves to 80+
Phase 2: Financial Reports	Week 5-6	Implement Balance Sheet, Income Statement, Period-close	Close financial reports working
Phase 3: Security Hardening	Week 7-8	Penetration testing, Security headers review, Access control	External pentest passes
Phase 4: UAT	Week 9-10	User acceptance testing, Real data migration, Training	Sign-off from business users
Phase 5: Go-Live	Week 11-12	Production deployment, Hypercare support, Monitoring	System stable for 2 weeks

Preconditions for Production Deployment:

- ☐ All 18 CRITICAL issues resolved and verified
- ☐ Security penetration test completed with no HIGH/Critical findings
- ☐ Balance Sheet and Income Statement reports operational
- ☐ Period-close mechanism tested and documented
- ☐ Stock integrity verified end-to-end (receipt -> storage -> delivery)
- ☐ Workflow SOD enforced and tested
- ☐ User training completed for all modules
- ☐ Data migration plan executed and validated
- ☐ Backup and recovery procedures tested
- ☐ Go-live checklist signed off by project sponsor

14. FINAL VERDICT & SIGN-OFF

Module Area	Score	Grade	Status
Sales Lifecycle	78/100	C+	CONDITIONAL PASS
Purchasing Cycle	76/100	C+	CONDITIONAL PASS
Accounting & Finance	68/100	D+	CONDITIONAL PASS
Security (OWASP)	62/100	D-	NEEDS ATTENTION
Inventory & Warehouse	62/100	D-	CONDITIONAL PASS
HR & Payroll	80/100	B-	CONDITIONAL PASS
Production & Mfg	62/100	D-	CONDITIONAL PASS
Workflow Engine	62/100	D-	CONDITIONAL PASS
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OVERALL AVERAGE	69/100	C+	CONDITIONAL PASS

AUDIT VERDICT: CONDITIONAL PASS

The HASSIBA Suite ERP demonstrates solid architectural foundations with comprehensive Algerian market localization and broad functional coverage across 47 modules. The system shows particular strength in HR/Payroll (80/100) where the Algerian tax engine is excellent, and in Sales/Purchasing lifecycles which have well-designed status workflows.

However, **18 CRITICAL issues** must be resolved before production deployment:

- 3 IDOR security vulnerabilities exposing sensitive data
- 5 stock integrity issues that could cause inventory discrepancies
- 3 missing financial statements required for business operations
- 3 workflow control gaps enabling fraud risk
- 4 other critical functional defects

With focused remediation over a **6-8 week period**, this system can achieve production-ready status with a target score of **85+/100**.

Algerian Localization Status: VERIFIED (85%)

[OK] NIF/NIS/RC/Al identifier support | [OK] SCF Chart of Accounts (Classes 1-8)

[OK] TVA (19%/9%/7%/0%), TAP, IRG, IBS taxes | [OK] CNAS/CASNOS social contributions

[OK] 58 Wilaya codes with communes | [OK] DZD currency throughout

[!!] RIB format validation needs strengthening | [!!] Arabic i18n not yet configured

This report represents a comprehensive audit opinion based on static analysis, runtime testing, and code

review. The audit team recommends addressing all CRITICAL and HIGH issues before proceeding to production deployment.

Lead Auditor	Technical Architect	Project Sponsor
Date: _____	Date: _____	Date: _____